

Wordlist:

1. Aerodynamics – The study of how air moves around objects, especially aircraft.
2. Lift – The upward force that allows an aircraft to rise into the air.
3. Thrust – The forward force produced by engines to propel an aircraft or spacecraft.
4. Drag – The resistance force acting opposite to the direction of motion through the air.
5. Weight – The downward force due to gravity acting on an aircraft.
6. Propulsion – The system or process that moves an aircraft or spacecraft forward.
7. Fuselage – The main body of an aircraft, holding crew, passengers, and cargo.
8. Wing – The part of an aircraft that generates lift.
9. Empennage – The tail section of an aircraft, providing stability and control.
10. Aileron – A hinged surface on the wing used to control roll.
11. Elevator – A control surface on the tail that adjusts pitch.
12. Rudder – A vertical control surface that controls yaw.
13. Angle of Attack (AOA) – The angle between the wing's chord line and the oncoming airflow.
14. Mach Number – The ratio of the speed of an object to the speed of sound.
15. Supersonic – Speeds greater than the speed of sound (Mach 1).
16. Hypersonic – Extremely high speeds, usually above Mach 5.
17. Turbine – A rotary engine part that extracts energy from fluid or gas flow.
18. Nozzle – A device that accelerates gases to produce thrust in jet engines and rockets.
19. Rocket Propulsion – The method of moving spacecraft using expulsion of gases.
20. Satellite – An object placed into orbit around Earth or another celestial body.
21. Orbit – The curved path of an object around a planet, star, or moon.
22. Trajectory – The flight path followed by an object through space.
23. Payload – The cargo carried by an aircraft or spacecraft, such as satellites or instruments.
24. Avionics – Electronic systems used in aircraft and spacecraft for navigation, communication, and control.
25. Flight Dynamics – The study of forces and moments that affect aircraft motion.
26. Stability – The ability of an aircraft to maintain controlled flight.
27. Control System – Mechanisms and software that govern the flight of an aircraft or spacecraft.
28. Spacecraft – A vehicle designed to travel in outer space.
29. Atmosphere – The layer of gases surrounding Earth that affects flight.
30. Re-entry – The process of a spacecraft returning to Earth's atmosphere from space.

30. Re-entry – The process of a spacecraft returning to Earth's atmosphere from space.

Task 1:

- a) The upward force that makes an aircraft rise. Lift
- b) The resisting force opposite to motion in the air. Drag
- c) The force of gravity acting downward. Weight
- d) The forward force that moves an aircraft ahead. Thrust
- e) The main body of an aircraft. Fuselage
- f) A surface that generates lift and supports the plane in the air. Wing
- g) A vertical control surface that changes yaw. Rudder
- h) A horizontal tail surface that controls pitch. Aileron
- i) The path an object follows through space. Trajectory
- j) The curved path around a planet, moon, or a star. Orbit

Task 2:

- 1. fuselage 6. Mach number
- 2. re-entry 7. Hypersonic
- 3. avionics 8. Payload
- 4. stability 9. Drag
- 5. Nozzle 10. Thrust

Task 3:

Aircraft parts:

fuselage, wing, rudder

Flight forces:

lift, drag, thrust, weight

Space & Orbit:

orbit, trajectory, spacecraft, satellite

Systems & Technology:

avionics, control system, propulsion, payload

Task 4:

Payload = cargo/load; emptiness (idk)

Trajectory = path; not existed

1. ... 2. ... 3. ... 4. ... 5. ... 6. ... 7. ... 8. ... 9. ... 10. ...

Trajectory = path; not existed
Avionics = aircraft electronics
Stability = balance, instability (better unsteadiness)
Lift = upward force, downward force (+weight)
Thrust = propulsion/drive, drag/resistance
Supersonic = super fast? subsonic (gozbgwobam)
Control system = controller; idk dead controller?

Task 5

- 1) The function of aileron is changing AOA or yaw.
It's necessary on take off and landing
- 2) Because wrong AOA will ruin take off or landing which is critical for flight.
Also they are responsible for keeping flight level.
- 3) Firstly they create intense aerodynamics heat due to huge air resistance. Secondly pilot must be very careful and qualified
- 4) They are taking fuel and oxidizer to produce hot, high pressure gasses.
Then gasses being accelerated by nozzle to generate thrust.
- 5) Because during re-entry gravity forces are getting stronger and stronger and due air resistance it causes overheating of the spacecraft.

Task 6:

An airplane flies because of the lift and thrust.
Wings is shaped to create lift and to keep flight level.
The main body, or fuselage is designed to hold people and cargo.
Rudder and elevator are made for pilots to manage a plane.

Launching a spacecraft into orbit is a complex process.
At first rocket propulsion systems produce thrust by igniting fuel.
That thrust must overcome Earth's gravity to get closer to orbit.
The rocket follows a certain trajectory making step by step a horizontal speed.
After the rocket stages separate, the spacecraft must accelerate an incredible speed $\approx 28,000 \text{ km/h}$ to achieve a stable orbit around Earth.

Airplanes and spacecrafts both are vehicle designed for travel but they operate in different environments.

Airplanes and spacecrafts both are vehicle designed for travel but they operate in different environments.

Airplanes generally rely on aerodynamics to fly in atmosphere. But spacecrafts rely on rocket propulsion to travel through the vacuum of space and achieve orbit. While jets need air for their engines and surfaces, others need must carry their own oxygen and use different control systems. Finally spacecraft has to carry about extreme heat of re-entry, a challenge airplanes do not face.